



University of Central Punjab
FOIT
Mid Term Examination

Program	BSCS	Semester	Spring 2025
Course Title	Linear Algebra	Course Code	CSSS2753
Time Allowed	1 hour 30 minutes	Total Marks	30
Name:		Registration #	

Instructions

1. Attempt all questions.
2. All your calculations should be on your answer sheet.

Question No.1 [CLO1] (1+1+1+1=4 Maks)

A tech startup is building a cloud-based task automation platform. The project is divided among three specialized teams: AI/ML Team, Cloud Infrastructure Team, and Security Team. The total number of developer hours allocated across all teams is 3600 hours. During a resource planning meeting, the following constraints and relationships between the teams' workloads were decided:

- a. The AI/ML Team requires 300 more hours than the combined hours of the Cloud Infrastructure and Security Teams due to the integration of complex learning models.
- b. The Cloud Infrastructure Team must be assigned twice the hours of the Security Team, as deployment and scalability tasks are significantly more demanding.
- c. The AI/ML Team's workload is also tied to system risk: it was determined that the difference between the AI/ML and Security Teams' hours must be equal to the number of hours assigned to the Cloud Infrastructure Team.

Formulate a system of three linear equations in three variables. Write an Augmented Matrix but DO NOT SOLVE THE SYSTEM.

Question No. 2 [CLO1] (5+1 = 6 Maks)

Find the solution by applying the Gauss-Jordan method. Explain each step and determine whether the system has a unique solution or infinite solutions

$$x + 2y + z = -1, \quad 6x + y + z = -4, \quad 2x - 3y - z = 0, \quad x - y = 1$$

Question No. 3 [CLO2] (2+2 = 4 Maks)

The equation of ellipse is given below. Find the new equation after rotating it through an angle of 90° , and sketch both the original and the rotated ellipse.

$$\frac{(x-3)^2}{16} + \frac{(y+2)^2}{9} = 1$$

Question No. 4 [CLO2] (3+3+2 = 8 Maks)

You are part of a 2D game development team working on a side-scrolling adventure game. Create the initial shape of the character using the following polygons:

Body (Rectangle): Vertices at (2.5, 1), (2.5, 4), (3.5, 4), (3.5, 1)

Left Arm (Rectangle): Vertices at (2.5, 2), (2.5, 3), (1.5, 2.5)

Find the new coordinates after applying reflection over the y-axis, and sketch both the original and the reflected character.

Question No. 5 [CLO2] (4+4 = 8 Maks)

As part of a secure messaging protocol in a cyber-defense training exercise, you are asked to **encrypt and decrypt** a short message applying the Hill Cipher method. Encode and decode the message "BUG HUNTER".

- Convert the message into numerical form using

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26=0

- Use the following key matrix for encryption

$$K = \begin{bmatrix} 3 & 3 \\ 1 & 6 \end{bmatrix}$$

Good Luck!